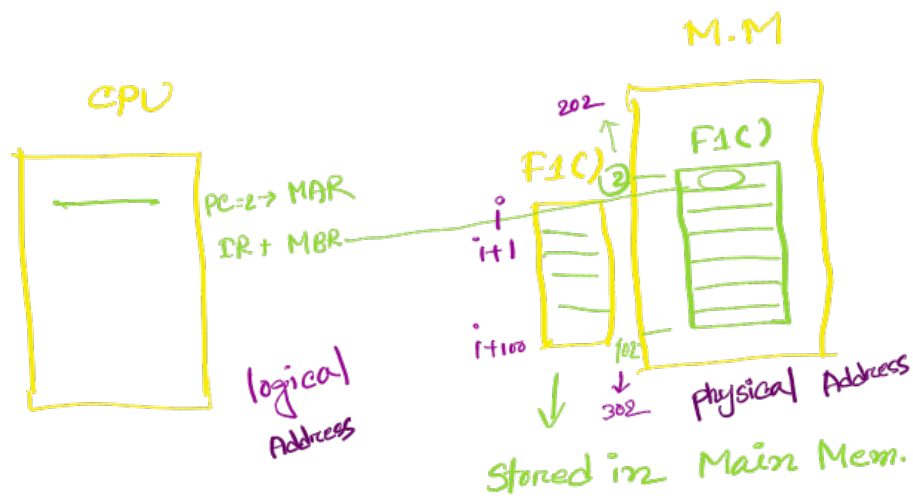
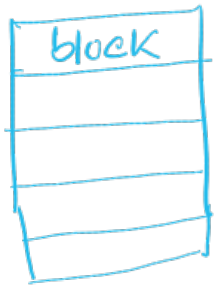
Cache Memory:



Dynamic Address.

FIFA 25
size - 85 GB
↓
Divided into sections

MMU
↓
Memory Management Unit.
logical → physical



32, 16, 18 gb.
PC

External storage stores it.
If there is not enough block
Swap in swap out.

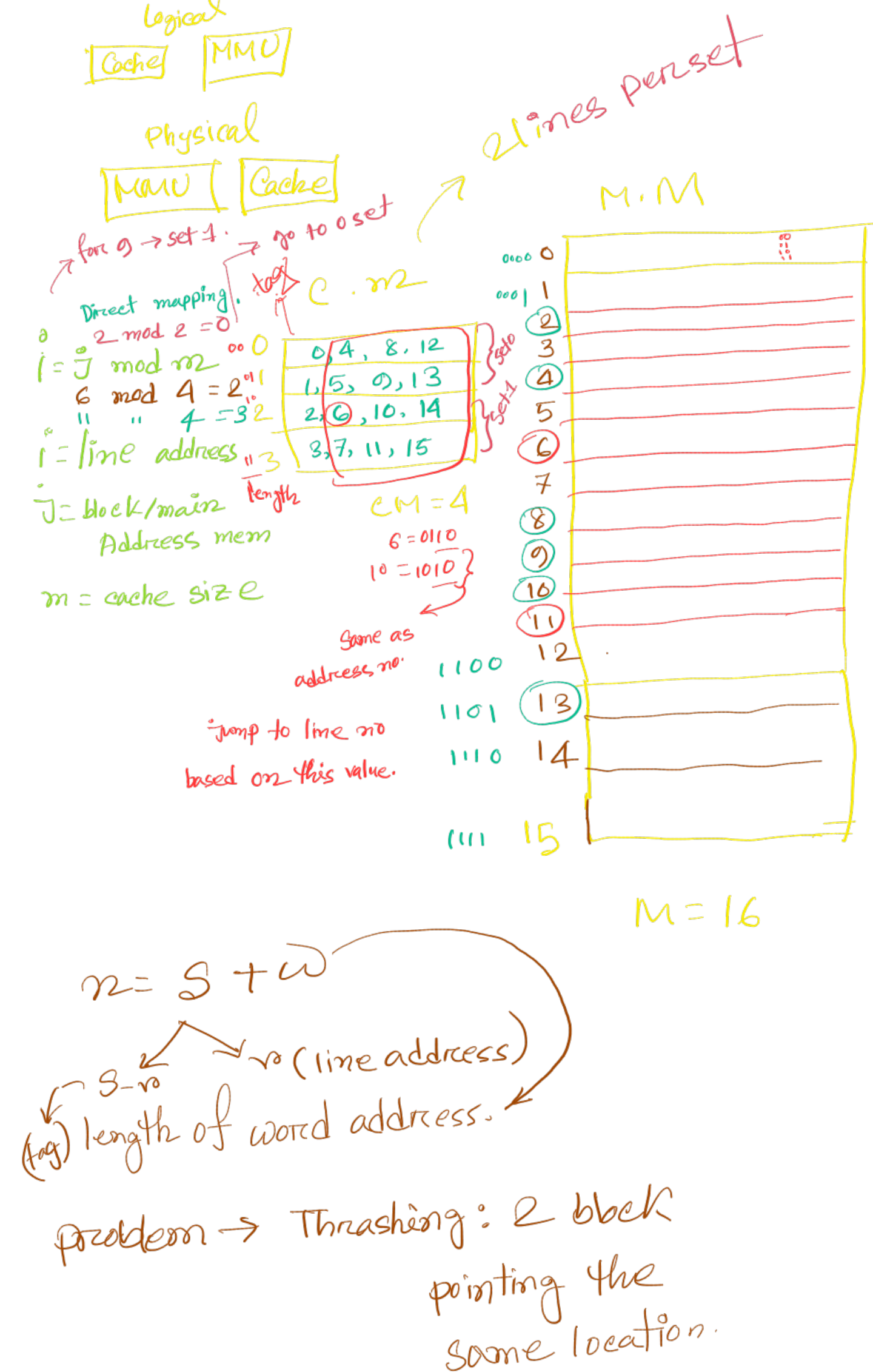
but we still need to add the
the previous block (in main memory) will get replaced
by the new one.

Virtual RAM
virtual/logical address.

Factory → cache size
Cache
logical
physical.

1 block from M.M
stored in cache line

Mapping


Logical
Cache
MMU
Physical
MMU
Cache
2 lines per set
M.M
for 0 → set 1.
go to 0 set
Direct mapping
tag
C . m
2 mod 2 = 0
i = j mod m
6 mod 4 = 2
11 " 4 = 3
i = line address
j = block/main Address mem
m = cache size
length
0, 4, 8, 12
1, 5, 9, 13
2, 6, 10, 14
3, 7, 11, 15
set0
set1
CM = 4
6 = 0110
10 = 1010
Same as address no.
jump to line no based on this value.
0000 0
0001 1
2
3
4
5
6
7
8
9
10
11
1100 12
13
1101
1110 14
1111 15
M = 16
n = S + w
s (line address)
S-r
(tag) length of word address.
problem → Thrashing: 2 block pointing the same location.

victim cache
L1 cache search 1st then L2 cache.

2 portion
→ word bit
→
→

set associative mapping.

set 0 → 0, 2, 4, 6, 8, ---- 14

set 1 → 1, 3, 5 ---- 15

search space reduced → just the set except the whole cache.

K, V, m

K = 1 (option) (direct mapping
(no of lines)





next week → Replacement Algorithms.